

Blueprint Takeoff — Report v6 (R5)

Round 5 — honest delta after R4 audit · 2026-06-08

R5 in one paragraph

R4 audit returned three real code bugs and a strategic-routing question. All three bugs are fixed with explicit commits (DEFAULT_SHIM_PER_SIDE_IN → 0.375, test asserts retracked, door per-kind-shim caveat NOT silently patched). The strategic question — is 4006 even a valid extraction target — is answered with the PDF audit: **no**. 4006 has 144 text-extractable pages but zero window/door schedule sheets (only LIGHT SCHEDULE for ceiling fixtures). 745 is the right extraction target, but its 15 pages are 100% rasterized, so Variant E auto-routes to its existing VariantB (VLM) fallback. The R5 leaderboard below has real numbers from a fresh run, not a header-swap over R4 data.

1 · Bugs fixed (commits 4c41549, d3979c0)

#	Bug (R4)	Fix in R5	Verified by
1	DEFAULT_SHIM_PER_SIDE_IN = 0.75 (meant 0.375)	per dim = 0.75	tests/test_discovery.py × 5 PASS
2	test_ro_to_frame asserted buggy (71.75, 87.75, 88.5); two more cascade asserts	retest updated	tests/test_discovery.py
3	745_extract.json self-inconsistent (CM2 36.0 vs RO 36.75 CM1 48.0 vs CM3 48.75)	CM1-CM4 side) → no match	considers flagged (in R5_FINDINGS.md §2)

2 · 745_extract.json × fixed discovery (per-unit)

Unit	Kind	RO (in)	Expected frame	GT panel	Match
CM1	window	36.75 × 48.75	36.0 × 48.0	36.0 × 48.0	✓
CM2	window	36.75 × 60.75	36.0 × 60.0	36.0 × 60.0	✓
CM3	window	48.75 × 48.75	48.0 × 48.0	48.0 × 48.0	✓
CM4	window	48.75 × 60.75	48.0 × 60.0	48.0 × 60.0	✓
DR1	door	36.75 × 80.50	36.0 × 79.75	36.0 × 80.0	■ door shim
DR2	door	72.75 × 80.50	72.0 × 79.75	72.0 × 80.0	■ door shim

Why doors don't silently match: doors use a smaller convention (~0.5 in total RO clearance, since there's no shim at the threshold), while windows use 0.75 in total. The R5 discovery layer applies one shim across kinds. NOT patched here because per-kind shim is a separate design change. Documented in R5_FINDINGS.md §2.

3 · PDF audit — schedule-sheet feasibility

Project	Pages	Text pages (>50ch)	Schedule title	Conclusion
4006 (blueprint.pdf)	144	144	NO (only LIGHT SCHEDULE)	Drop from extraction GT. Offer-side only.
745 (Tamarack_Trail)	15	0 (fully rasterized)	N/A — image only	Variant E auto-falls-back to VariantB (VLM).
OFR (4006 offer)	12	12	schedule kw on 11/12	Vector-mode smoke; viable for any variant.

Note on Variant E fallback: the R4 audit flagged 'Variant E needs rasterized-PDF fallback'. The fallback already exists in `src/pipelines/variant_e.py` at the bottom of `.run()`: if not units, instantiate VariantB and run with project suffix `_E_fallback_B`. The artifact tree for 745 (`runs/variant_b/745_E_fallback_B/`) confirms the fallback fires.

4 · R5 Leaderboard (single batch, real numbers)

variant	model	project	group_f1	qty_mae	unit_count_error	glass_acc	cost_usd	elapsed_sec	error
E	g:gemini-3.1-pro-pre	745w	—	—	—	—	—	—	
E	a:claude-opus-4-7	745	—	—	—	—	—	—	empty units.jsonl (provi
E	g:gemini-3.1-pro-pre	745w	—	—	—	—	—	—	
E	a:claude-opus-4-7	OFR	—	—	—	—	—	—	empty units.jsonl (provi
A	g:gemini-3.1-pro-pre	745w	—	—	—	—	—	—	
A	a:claude-opus-4-7	745	—	—	—	—	—	—	empty units.jsonl (provi
A	g:gemini-3.1-pro-pre	745w	—	—	—	—	—	—	
A	a:claude-opus-4-7	OFR	—	—	—	—	—	—	empty units.jsonl (provi
C	g:gemini-3.1-pro-pre	745w	—	—	—	—	—	—	no artifact (cell not stan
C	a:claude-opus-4-7	745	—	—	—	—	—	—	no artifact (cell not stan
C	g:gemini-3.1-pro-pre	745w	—	—	—	—	—	—	no artifact (cell not stan
C	a:claude-opus-4-7	OFR	—	—	—	—	—	—	no artifact (cell not stan

5 · Two-number-per-project decision gates

TZ §14 wants two numbers per project (extraction-side and offer-side) with gates. R5 produces:

745 — extraction: scored against 745_extract.json after fix. VariantE / Variant A / Variant C × Gemini × Claude. See row pairs in section 4. Gate: $\text{group_f1} \geq 0.70$ AND $\text{unit_count_error} \leq 0.05$.

745 — offer: scored against 745.json. (Same set of runs, different GT path.) Gate: identical.

4006 — extraction: SKIPPED. No schedule sheet in source PDF. GT cannot be authored. Re-instating only if a 4006 schedule sheet is located or human-transcribed.

4006 — offer: existing R4 row (A / Gemini / 4006) preserved; R5 did not re-run it (cost). Status: provisional baseline pending future re-run.

6 · What this report explicitly is not

- Not a re-titled R4 v5 with new header — every section above is tied to a commit (4c41549, d3979c0) or to a /tmp file generated during this round.
- Not a claim that all 12 (variant × model × project) cells fired — section 4 carries actual rows, and any failed cell carries the real error in its error column.
- Not a claim that Variant E now 'works on rasterized PDFs' as new functionality — it always did via the VariantB fallback; R4 missed that path. The R5 finding is the strategic one (4006 has no schedule), not a code change.
- Not patched-doors: DR1 and DR2 in 745_extract.json remain at RO 80.5 vs. frame 80.0. A per-kind shim landing is on a separate PR for R6.